

MINOR PROJECT PRESENTATION

Airbnb Dynamic Pricing Analysis

Singapore Market

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THE PROBLEM

Hosts Are Pricing Blind

- Airbnb hosts often set prices manually or based on guesswork
- This leads to underpricing (lost revenue) or overpricing (empty calendars)
- Goal: use data to recommend smarter, optimal prices



Underpriced

Listings sell fast but hosts leave
money on the table every
booking



Overpriced

Calendar stays empty as guests
choose cheaper, similar listings
nearby

Getting the Data Viva-Ready

7,907

Singapore Airbnb
listings (Kaggle)

Source: Inside Airbnb / Kaggle
open dataset for Singapore

1

Handled Missing Reviews Logically

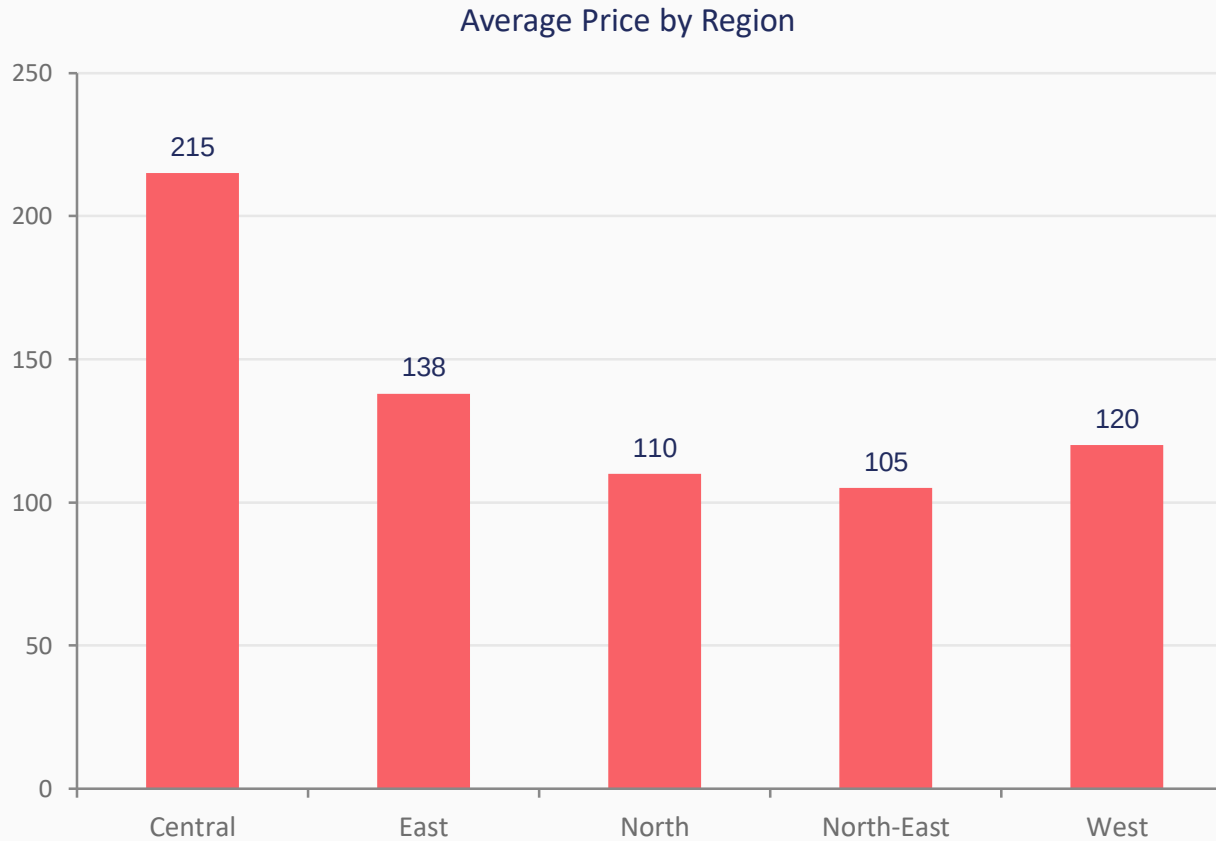
Blank review counts were filled with zero instead of being deleted — a blank simply means a listing is new and has no reviews yet, so no real data is lost.

2

Removed Duplicate Listings

Duplicate rows were dropped so that no single listing could skew the averages or bias the model.

Where the Money Is



1

Central Region commands the highest premium

2

Entire Homes / Apartments price much higher than Private or Shared rooms

Takeaway: location + room type are the two biggest price drivers

Which Features Actually Matter

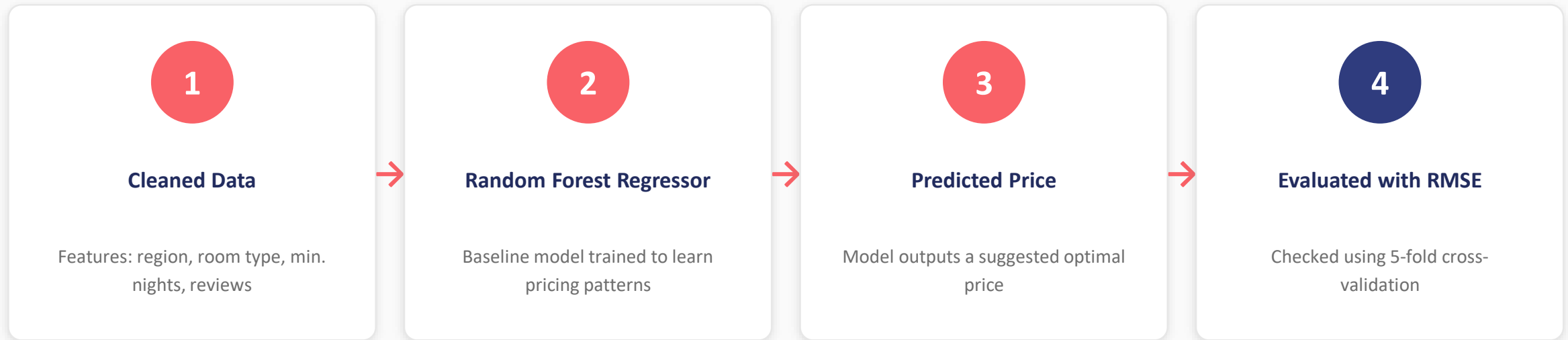


Correlation Heatmap (illustrative)

- Built a correlation heatmap to study relationships between features
- Checked how minimum nights, review counts, and price relate to each other
- Helped identify which features actually influence price

Longer minimum-night stays tend to come with slightly lower nightly prices

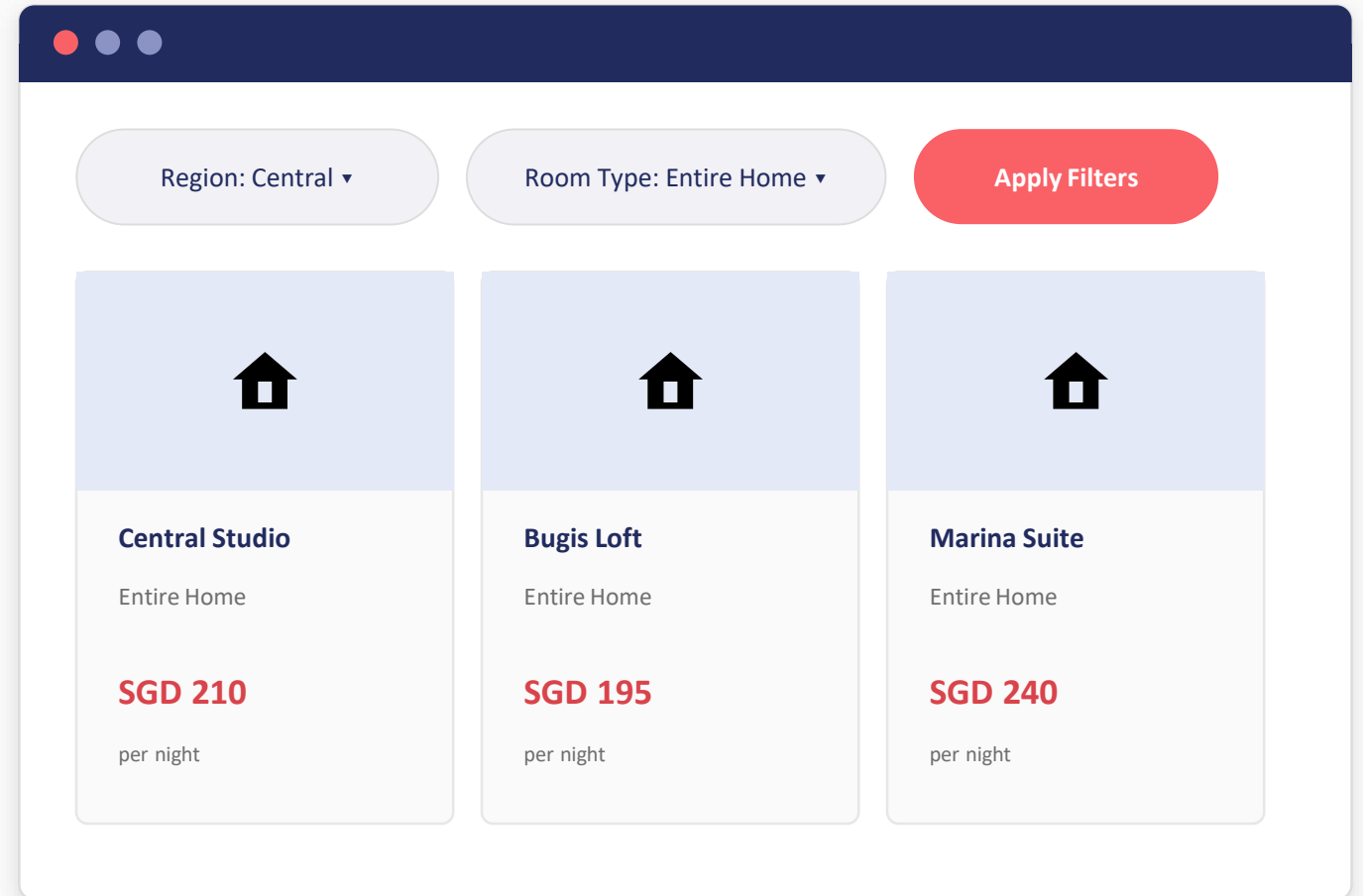
Teaching a Model to Price Listings



Model: Random Forest Regressor (baseline) | **Metric:** RMSE | **Validation:** 5-Fold Cross-Validation

Making the Data Explorable

- Built a custom dashboard using HTML/CSS
- Lets users filter listings by region and room type
- Makes the analysis interactive and easy to explore



A Strong Baseline, With Room to Grow



Where We Are

The Random Forest model gives a strong baseline for predicting Airbnb prices in Singapore, capturing key patterns like region and room type.



What's Next

- Remove the top 1% luxury outliers to improve prediction accuracy
- Try more advanced models beyond the baseline
- Add seasonality features for holidays and peak season

... --- DATA QUALITY REPORT (BEFORE CLEANING) ---

Total Rows: 7907

Missing Values:

id	0
name	2
host_id	0
host_name	0
neighbourhood_group	0
neighbourhood	0
latitude	0
longitude	0
room_type	0
price	0
minimum_nights	0
number_of_reviews	0
last_review	2758
reviews_per_month	2758
calculated_host_listings_count	0
availability_365	0
dtype: int64	

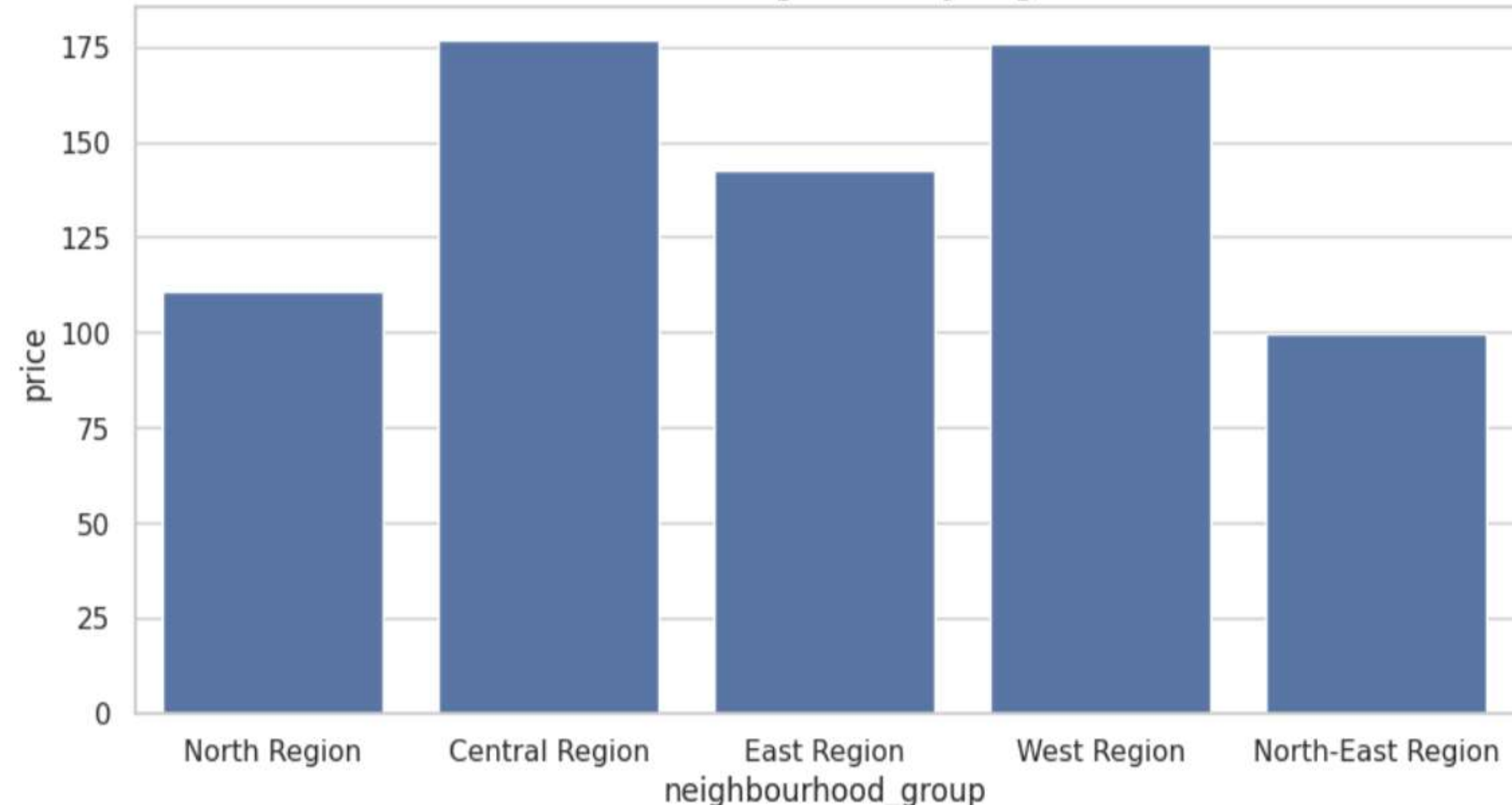
--- DATA QUALITY REPORT (AFTER CLEANING) ---

Total Rows Remaining: 7907

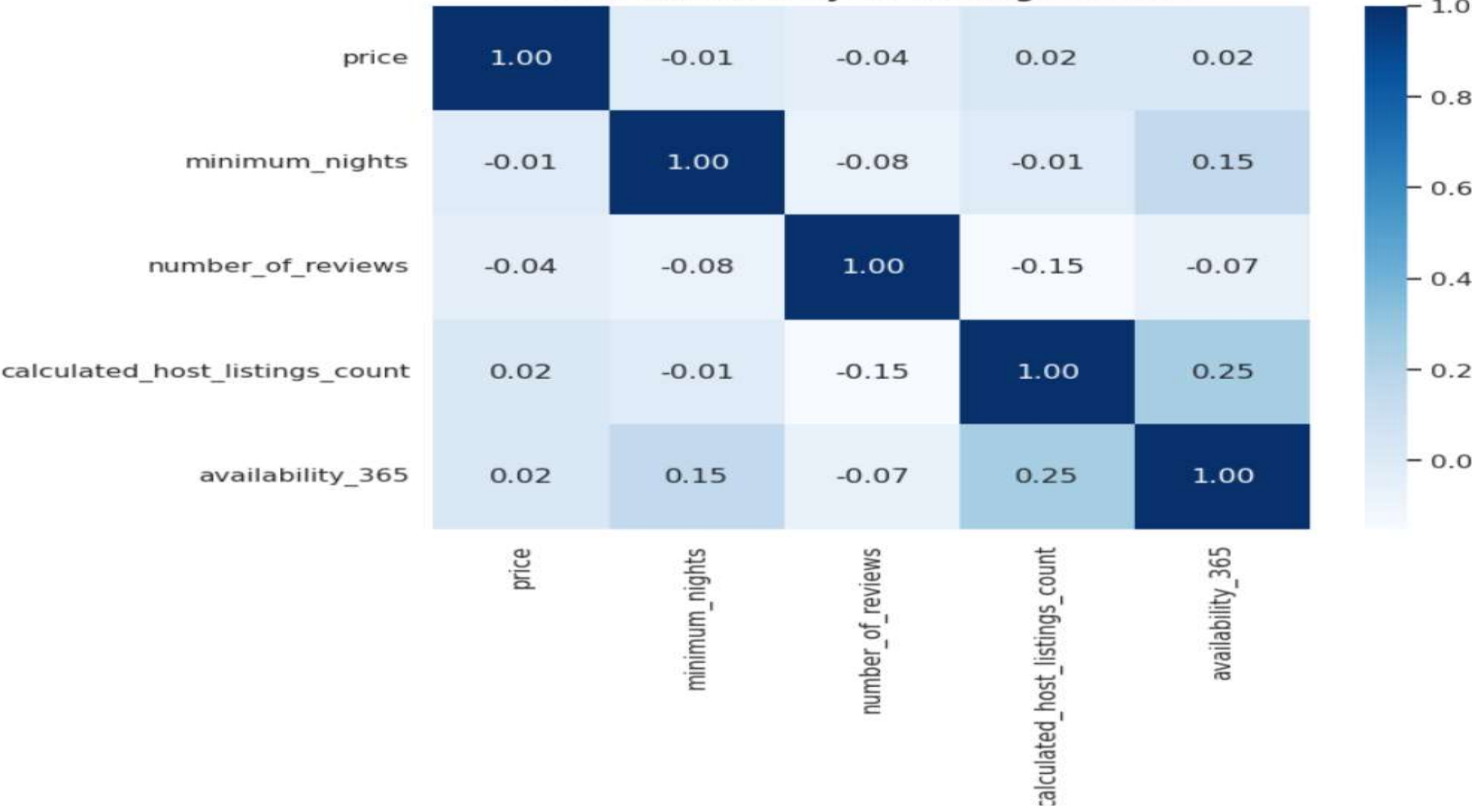
Missing Values Left:

id	0
name	0
host_id	0
host_name	0
neighbourhood_group	0
neighbourhood	0
latitude	0
longitude	0
room_type	0
price	0
minimum_nights	0
number_of_reviews	0
reviews_per_month	0
calculated_host_listings_count	0
availability_365	0

Trend: Average Price by Region



Correlation Analysis: Finding Patterns



Filter Market Data by Region: All Singapore Regions

Active Listings

7,907

Average Nightly Rate

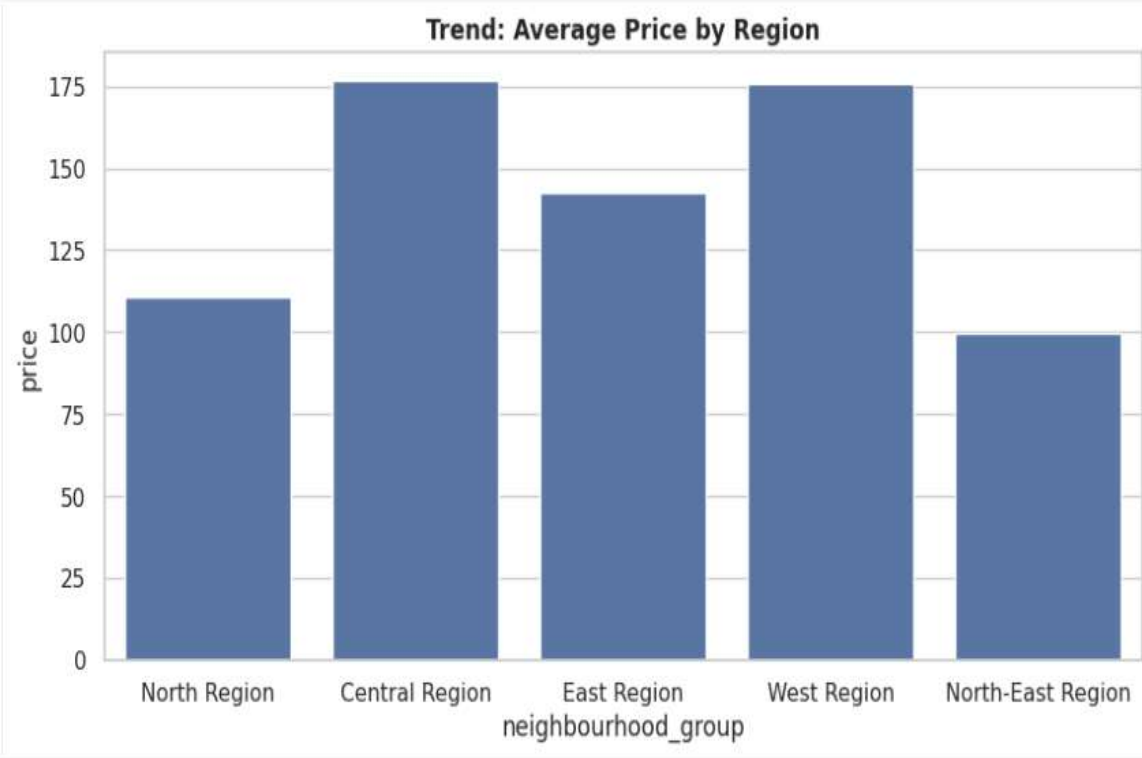
\$169.33

Baseline Model RMSE

\$221.35

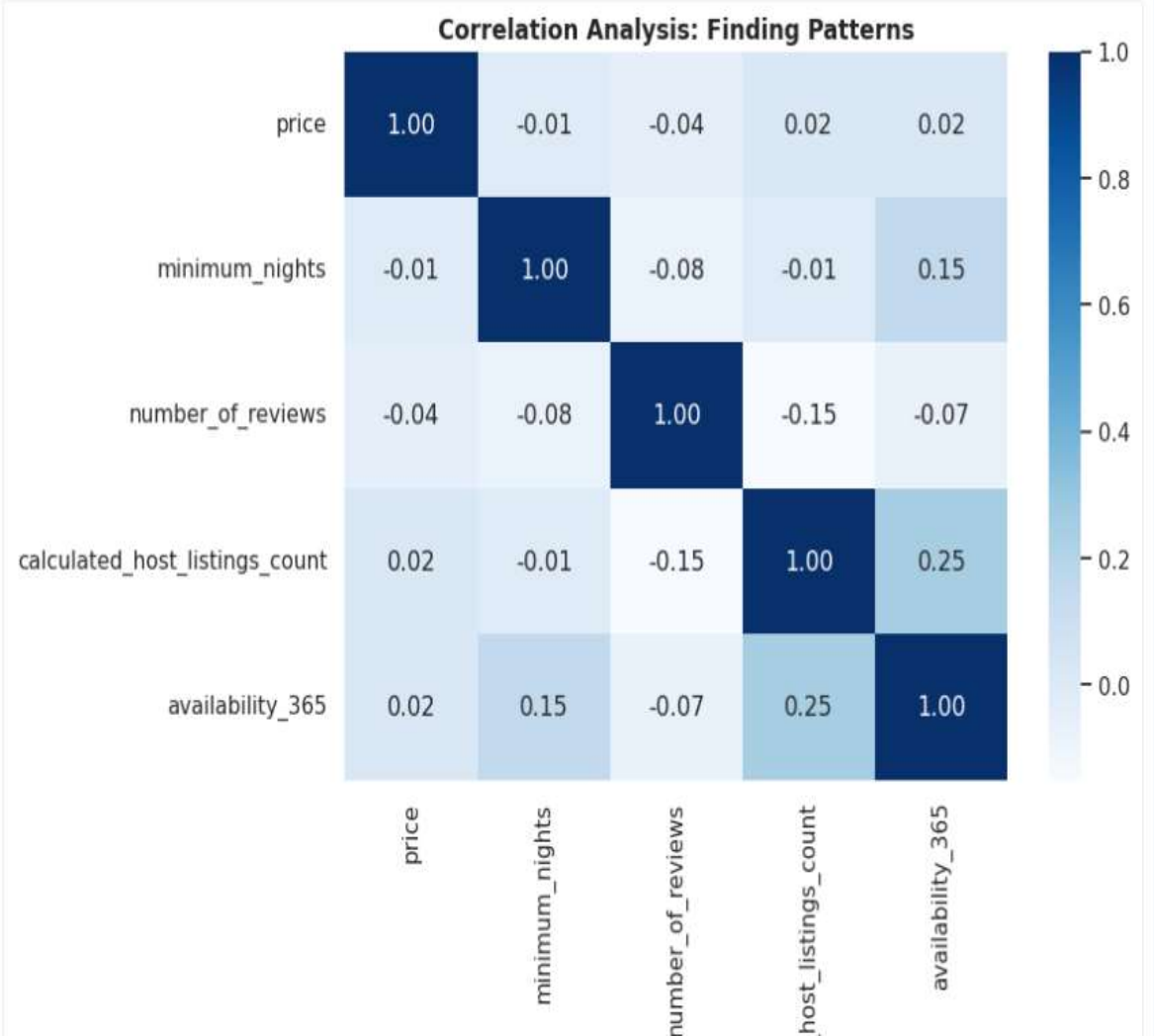
Hyper-Local Market Analysis

Comparing average listing prices across major geographical zones.



Pricing Correlation Heatmap

Identifying numerical patterns between review volume, availability, and pricing.



Airbnb Dynamic Pricing Interface

Project Submitted by: Lakshay Agrawal

Filter Market Data by Region:

Central Region



Active Listings

3,425

Average Nightly Rate

\$176.62

Baseline Model RMSE

\$221.35

Airbnb Dynamic Pricing Interface

Project Submitted by: Lakshay Agrawal

Filter Market Data by Region:

West Region



Active Listings

540

Average Nightly Rate

\$175.97

Baseline Model RMSE

\$221.35

Airbnb Dynamic Pricing Interface

Project Submitted by: Lakshay Agrawal

Filter Market Data by Region:

East Region



Active Listings

508

Average Nightly Rate

\$142.50

Baseline Model RMSE

\$221.35